Science Bulletin 68 (2023) 2106–2114

Contents lists available at ScienceDirect

Science Bulletin

journal homepage: www.elsevier.com/locate/scib

Article

A life-threatening bleeding prediction model for immune thrombocytopenia based on personalized machine learning: a nationwide prospective cohort study

Zhuo-Yu An [a,b,c,d,1], Ye-Jun Wu [a,b,c,d,1], Yu Hou [e,1], Heng Mei [f,1], Wei-Xia Nong [g,1], Wen-Qian Li [h,1], Hu Zhou [i], Ru Feng [j], Jian-Ping Shen [k], Jun Peng [e], Hai Zhou [e], Yi Liu [l], Yong-Ping Song [i], Lin-Hua Yang [m], Mei-Yun Fang [n], Jian-Yong Li [o], Yun-Feng Cheng [p], Peng Liu [q], Ya-Jing Xu [r], Zhao Wang [s], Yi Luo [t], Zhen Cai [u], Hui Liu [j], Jing-Wen Wang [t], Juan Li [u], Xi Zhang [v], Zi-Min Sun [w], Xiao-Yu Zhu [w], Xin Wang [x], Rong Fu [y], Liang Huang [z], Shao-Yuan Wang [aa], Tong-Hua Yang [ab], Li-Ping Su [ac], Liang-Ming Ma [ad], Xie-Qun Chen [ae], Dai-Hong Liu [af], Hong-Xia Yao [ag], Jia Feng [ah], Hong-Yu Zhang [ah], Ming Jiang [ai], Ze-Ping Zhou [aj], Wen-Sheng Wang [ak], Xu-Liang Shen [al], Yangjin Baima [am], Yue-Ying Li [an], Qian-Fei Wang [an], Qiu-Sha Huang [a,b,c,d], Hai-Xia Fu [a,b,c,d], Xiao-Lu Zhu [a,b,c,d], Yun He [a,b,c,d], Qian Jiang [a,b,c,d], Hao Jiang [a,b,c,d], Jin Lu [a,b,c,d], Xiang-Yu Zhao [a,b,c,d], Ying-Jun Chang [a,b,c,d], Tao Wu [ao], Yao-Zhu Pan [ao], Lin Qiu [ap], Da Gao [aq], A-Rong Jin [ar], Wei Li [as], Su-Jun Gao [as], Lei Zhang [at,*], Ming Hou [e,*], Xiao-Jun Huang [a,b,c,d,*], Xiao-Hui Zhang [a,b,c,d,*], on behalf of the National Cooperative ITP Working Group

[a] *Peking University People's Hospital, Peking University Institute of Hematology, Beijing 100044, China*
[b] *Collaborative Innovation Center of Hematology, Peking University, Beijing 100044, China*
[c] *Beijing Key Laboratory of Hematopoietic Stem Cell Transplantation, Beijing 100044, China*
[d] *National Clinical Research Center for Hematologic Disease, Beijing 100044, China*
[e] *Department of Hematology, Qilu Hospital, Cheeloo College of Medicine, Shandong University, Jinan 250014, China*
[f] *Institute of Hematology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430074, China*
[g] *Department of Hematology, First Affiliated Hospital, School of Medicine, Shihezi University, Shihezi 832002, China*
[h] *Department of Hematology, Qinghai Provincial People's Hospital, Xining 810007, China*
[i] *Henan Cancer Hospital, Affiliated Cancer Hospital of Zhengzhou University, Henan Institute of Hematology, Zhengzhou 450008, China*
[j] *Department of Hematology, Beijing Hospital, Beijing 100044, China*
[k] *Department of Hematology, Zhejiang Provincial Hospital of Traditional Chinese Medicine, Hangzhou 310006, China*
[l] *Department of Hematology, Senior Department of Hematology, the Fifth Medical Center of PLA General Hospital, Beijing 100044, China*
[m] *Department of Hematology, the Second Hospital of Shanxi Medical University, Taiyuan 030001, China*
[n] *Department of Hematology, Affiliated Zhongshan Hospital of Dalian University, Dalian 116001, China*
[o] *Department of Hematology, The First Affiliated Hospital with Nanjing Medical University (Jiangsu Province Hospital), Nanjing 210029, China*
[p] *Department of Hematology, Zhongshan Hospital, Fudan University, Shanghai 200032, China*
[q] *Department of Hematology, Xiangya Hospital, Central South University, Changsha 410008, China*
[r] *Department of Hematology, Beijing Friendship Hospital, Capital Medical University, Beijing 100044, China*
[s] *Institute of Hematology, the First Affiliated Hospital, Zhejiang University, Hangzhou 310058, China*
[t] *Department of Hematology, Beijing Tongren Hospital, Beijing 100005, China*
[u] *Department of Hematology, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou 510080, China*
[v] *Medical Center of Hematology, Xinqiao Hospital, Army Medical University, Chongqing 400037, China*
[w] *Department of Hematology, First Affiliated Hospital of USTC, Division of Life Sciences and Medicine, University of Science and Technology of China, Hefei 230026, China*
[x] *Department of Hematology, Shandong Provincial Hospital, Cheeloo College of Medicine, Shandong University, Jinan 250014, China*
[y] *Department of Hematology, Tianjin Medical University General Hospital, Tianjin 300052, China*
[z] *Institute of Hematology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430074, China*
[aa] *Department of Hematology, Fujian Institute of Hematology, Fujian Medical University Union Hospital, Fuzhou 350001, China*
[ab] *Department of Hematology, the First People's Hospital of Yunnan Province, Kunming 650032, China*
[ac] *Department of Hematology, Shanxi Tumor Hospital Affiliated to Shanxi Medical University, Taiyuan 030001, China*
[ad] *Department of Hematology, Shanxi Bethune Hospital of Shanxi Medical University, Taiyuan 030001, China*
[ae] *Department of Hematology, The Affiliated Hospital of Northwest Hospital, Xi'an No.3 Hospital, Xi'an 710054, China*
[af] *Department of Hematology, Chinese PLA General Hospital & PLA Medical School, Beijing 100044, China*
[ag] *Department of Hematology, Hainan Affiliated Hospital of Hainan Medical University, Hainan General Hospital, Haikou 570311, China*
[ah] *Department of Hematology, Peking University Shenzhen Hospital, Shenzhen 518036, China*

* Corresponding authors.
E-mail addresses: zhanglei1@ihcams.ac.cn (L. Zhang), houming2019@tom.com (M. Hou), xjhrm@medmail.com.cn (X.-J. Huang), zhangxh@bjmu.edu.cn (X.-H. Zhang).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.scib.2023.08.001
2095-9273/© 2023 Science China Press. Published by Elsevier B.V. and Science China Press.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

[ai] Center of Hematologic Diseases, The First Affiliated Hospital of Xinjiang Medical University, Urumchi 830054, China
[aj] Department of Hematology, The Second Affiliated Hospital of Kunming Medical University, Kunming 650500, China
[ak] Department of Hematology, Peking University First Hospital, Beijing 100034, China
[al] Department of Hematology, Heping Hospital Affiliated to Changzhi Medical College, Changzhi 046000, China
[am] Department of Hematology, People's Hospital of Tibet Autonomous Region, Lhasa 850000, China
[an] Key Laboratory of Genomic and Precision Medicine, Collaborative Innovation Center of Genetics and Development, Beijing Institute of Genomics, Chinese Academy of Sciences, Beijing 100029, China
[ao] Department of Hematology, The 940th Hospital of Joint Logistics Support Force of Chinese PLA, Lanzhou 730050, China
[ap] Institute of Hematology, Harbin the First Hospital, Harbin 150001, China
[aq] Department of Hematology, The Affiliated Hospital of Inner Mongolia Medical University, Hohhot 010050, China
[ar] Department of Hematology, Inner Mongolia People's Hospital, Hohhot 010017, China
[as] Department of Hematology, The First Bethune Hospital of Jilin University, Changchun 130021, China
[at] State Key Laboratory of Experimental Hematology, Institute of Hematology and Blood Disease Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Tianjin 300020, China

ARTICLE INFO

Article history:
Received 30 November 2022
Received in revised form 24 February 2023
Accepted 26 June 2023
Available online 3 August 2023

Keywords:
Critical bleeding
Severe bleeding
Immune thrombocytopenia
Machine learning
Prediction model

ABSTRACT

Rare but critical bleeding events in primary immune thrombocytopenia (ITP) present life-threatening complications in patients with ITP, which severely affect their prognosis, quality of life, and treatment decisions. Although several studies have investigated the risk factors related to critical bleeding in ITP, large sample size data, consistent definitions, large-scale multicenter findings, and prediction models for critical bleeding events in patients with ITP are unavailable. For the first time, in this study, we applied the newly proposed critical ITP bleeding criteria by the International Society on Thrombosis and Hemostasis for large sample size data and developed the first machine learning (ML)-based online application for predict critical ITP bleeding. In this research, we developed and externally tested an ML-based model for determining the risk of critical bleeding events in patients with ITP using large multicenter data across China. Retrospective data from 8 medical centers across the country were obtained for model development and prospectively tested in 39 medical centers across the country over a year. This system exhibited good predictive capabilities for training, validation, and test datasets. This convenient web-based tool based on a novel algorithm can rapidly identify the bleeding risk profile of patients with ITP and facilitate clinical decision-making and reduce the occurrence of adversities.

© 2023 Science China Press. Published by Elsevier B.V. and Science China Press. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Primary immune thrombocytopenia (ITP) is a commonly acquired autoimmune disease characterized by reduced platelet (PLT) production and increased PLT destruction. Patients with ITP have a low PLT count and a high risk of bleeding. While most ITP patients typically experience a positive prognosis, rare but critical ITP bleeding events lead to life-threatening complications for the patients, severely affecting their prognosis, quality of life, and treatment decisions [1–4]. Several predictors of critical ITP bleeding, such as low PLT count and previous bleeding history, have been previously reported [5–8]. Moreover, response to PLT thrombopoietin receptor agonist therapy was identified as a protective factor for critical ITP bleeding [9–13].

Our previous study identified risk stratification and outcomes for intracranial hemorrhage (ICH) in patients with ITP. However, data on critical ITP bleeding, such as ICH, gastrointestinal bleeding, and respiratory bleeding, are limited [14]. To date, large sample size data, consistent definitions, large-scale multicenter findings, and prediction models for critical ITP bleeding events in patients with ITP are unavailable.

The recent development of clinical prediction models, particularly those that employ machine learning (ML) methods, has provided powerful tools for precision diagnosis and early intervention of diseases. The ML approaches can overcome some limitations of current risk-prediction analysis methods by employing computer algorithms for processing large datasets with numerous multidimensional variables. The ML approaches identify high-dimensional nonlinear relationships between clinical features, which produce reliable data that can predict outcomes. Various ML methods have been widely applied to clinical prediction models. Among them, random forest (RF) and extreme gradient boosting (XGBoost) are ensemble algorithms that are efficient, accurate, and suitable for large-scale and high-dimensional data. Logistic regression is a simple and easy-to-understand linear model, while the light gradient-boosting machine (LightGBM) has excellent performance in handling large-scale data [15]. Nevertheless, there is an unmet need for personalized patient-management strategies and an urgent need for effective tools for predicting the risk of critical bleeding in patients with ITP [5].

Here, we integrated clinical and laboratory data from a nationwide multicenter study to develop a clinical prediction model. Furthermore, we performed retrospective validation and prospective testing of the models with large sample sizes to improve their robustness and utility. Our analyses showed that the developed tools were simple and convenient and could quickly assess newly admitted patients with ITP to achieve early identification of high-risk, life-threatening bleeding events, thus reducing disability and mortality rates.

2. Materials and methods

2.1. Patients and data sources

This study was approved by the Institutional Review Board (IRB) of a leading center (2022PHB064-001) and was conducted in compliance with the *Declaration of Helsinki*. Because this was an observational and noninvasive study, informed consent was waived by the IRB. All obtained data were kept confidential and anonymous to protect the privacy and rights of the participants.

We first conducted a multicenter, retrospective cohort study that involved the consecutive, unselected inclusion of patients diagnosed with ITP. These individuals were admitted to eight

medical centers in China from December 2002 to September 2019 using electronic medical records. The data of patients with ITP for model development, validation, and testing were collected from the centers by the National ITP Collaborative Working Group. After the development and retrospective validation of the model, we established prospective observational cohorts of 39 medical centers across China from November 2021 to November 2022 to further assess the efficacy of the model (Table S1 online). The prospective cohort was registered at ClinicalTrials.gov (NCT05116423) and included patients with ITP who were continuously and non-selectively included from both outpatient and inpatient departments of each center. Their relevant information was collected through electronic medical records. A follow-up for a minimum of six months was conducted, during which further bleeding data collection was performed if critical ITP bleeding occurred. Patients with primary ITP aged ≥18 years were included, and those with a diagnosis of rheumatic or connective disease, cancer (solid tumors or leukemia), or primary immunodeficiency were excluded. Patients currently experiencing critical ITP bleeding were also excluded, as defined in the following section. This study used the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis Statement as the reporting guideline.

2.2. Definitions

Patients with primary ITP were diagnosed, by following the 2019 American Society of Hematology guidelines, in retrospective and prospective cohorts, and patients with secondary ITP were excluded [3,16]. Inconsistency in the definition of critical ITP bleeding was observed in previous studies because of inherent differences in the bleeding measurement tools. In our study, we adopted the criteria proposed by the Platelet Immunology Scientific Standardization Committee of the International Society on Thrombosis and Hemostasis (ISTH) [17]: (1) anatomical: bleeding occurring in an anatomical site that involves intracranial, intraspinal, intraocular, retroperitoneal, pericardial, or intramuscular compartment syndrome or (2) physiological: ongoing bleeding leading to hemodynamic instability or respiratory compromise.

Hemodynamic instability was defined as bleeding with a 20-mmHg (1 mmHg = 0.133 kPa) decrease in systolic blood pressure from baseline, a 10-mmHg decrease in mean arterial pressure, or blood pressure <90 mmHg on admission. Furthermore, bleeding leading to respiratory compromise was defined by a respiratory rate ≥30/min, an oxygen saturation ≤92%, a ratio of arterial partial pressure of oxygen to fraction of oxygen absorbed <300, or the need for invasive or noninvasive mechanical ventilation [18–20].

2.3. Feature selection for modeling

Candidate predictors were selected from routinely available admission data and those compiled from previous retrospective studies, expert opinions, consensus statements, and previous observational cohorts, collected from the electronic medical records of the cohorts and confirmed by two hematologists in each center. Specifically, for patients enrolled in the prospective cohorts, the assessment was performed every four weeks to obtain medical information. Telephone contact was made if the information was incomplete.

In the retrospective and prospective cohorts, information closely related to the clinical characteristics of ITP was collected. The variables included available patient sociodemographic data (e.g., age and gender), ITP diagnosis (e.g., new onset, persistent, and chronic), comorbidities (e.g., cardiovascular disease and uncontrolled diabetes), signs and symptoms (e.g., purpura, petechiae, and hemoptysis), and laboratory tests (e.g., PLT count and lymphocyte count) [14,17,21,22]. The laboratory test results include the most recent test results before the evaluation and the test results obtained at diagnosis. All steps of data cleaning and processing were directed by statisticians and confirmed by at least two researchers to ensure the authenticity and reliability of the statistical results.

After data cleaning, sampling, and preprocessing, several variables remained. For the retrospective cohort, features with >20% missing values were excluded. Meanwhile, the prospective cohort had no features with missing value proportions exceeding 20%. The multiple imputation method was used to supplement the data with a missing proportion of <20% in the dataset. To rank the importance of the candidate variables, importance analysis was performed through Lasso regression. To ensure a robust and reliable model, we selected the top ten features for model construction.

2.4. Development, evaluation, and validation of the models

Overall, the multicenter retrospective cohort was randomly split into a training dataset (1494 patients, 71%) and an internal validation dataset (600 patients, 29%). The prospective cohort (1097 patients) served to test the efficacy of the model. Hyperparameter tuning and cross-validation analysis were performed during validation. This analysis is robust for model evaluation prior to retrospective testing on separate datasets and maximizes the potential performance of ML models.

Seven classical ML models, namely logistic regression, LightGBM, adaptive boosting, multilayer perceptron, support vector machine, RF, and XGBoost [23], were developed to predict critical ITP bleeding. Furthermore, we selected RF, XGBoost, LogisticBoost, and LightGBM, which performed better and optimized their parameters for statistical comparison in the final analysis.

We used the area under the receiver operating characteristic curve (AUC), calibration curve, and decision curve to assess model performance [24]. The C statistic assesses the overall capability of the model to distinguish between high- and low-risk patients, the calibration curve shows the relationship between the predicted and actual outcomes, and the decision curve analysis (DCA) visually quantifies the net benefit of each model, which is shown as a value for each model. The ML model was implemented in the open-source Python 3X and Project Jupyter version 1.2.3 (Anaconda, Inc., Austin, USA).

2.5. Statistical analysis

Categorical variables were reported as counts (percentages) and compared through the chi-square test. The distribution of continuous data was defined based on the Kolmogorov–Smirnov test; variables with non-normal distribution were presented as median (Q1, Q3). Independent *t*-test or analysis of variance was used to test the differences between two groups of continuous variables with a normal distribution. The Mann-Whitney *U* test was used for the differences between two groups of continuous variables without normal distribution. Furthermore, the relative risk was estimated by logistic regression analysis using the odds ratio and its 95% confidence interval. Data were analyzed using IBM SPSS 24.0 (IBM Corp., Armonk, USA), R version 4.0.0 (R Foundation for Statistical Computing, Vienna, Austria), and Python (Python Software Foundation, Wilmington, USA) software. All the *P* values are two-sided, and $P < 0.05$ was considered statistically significant.

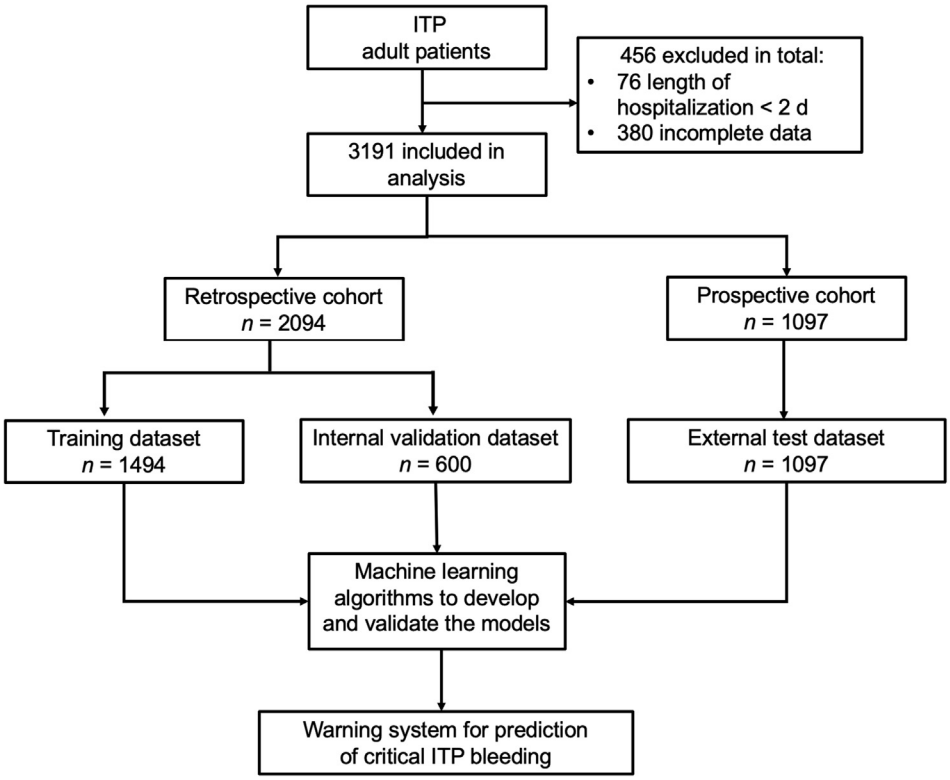

Fig. 1. Flow diagram of the study design.

3. Results

3.1. Patient characteristics

The retrospective cohort involved 2094 eligible adult patients with ITP admitted to eight medical centers in China from December 2002 to September 2019 (Fig. 1), and the cohort was divided into training (1494 patients) and internal validation (600 patients) datasets. The retrospective cohort (2094 patients) had a median age of 53.0 years (Q1, Q3 = 36, 68 years) and a female-to-male ratio of 1.66:1. A total of 597 patients with ITP in the retrospective cohort experienced critical bleeding events (Table 1). The prospective cohort involved a total of 1097 patients from 39 medical centers located nationwide (Fig. 1), which included a total of 329 hospitalized patients and 768 outpatients. Among 50 patients who experienced critical bleeding in the prospective cohort, 36 hospitalized patients experienced critical bleeding after enrollment, while 14 outpatients experienced critical bleeding after enrollment.

3.2. Bleeding profiles

The incidence of critical ITP bleeding in the patients with ITP in the retrospective cohort was 28.24% (Fig. S1a online). Each patient was diagnosed with a specific major type of bleeding according to the critical bleeding diagnostic criteria. The most frequent type was intracranial bleeding (54.46%), followed by respiratory compromise (20.66%) and hemodynamic instability bleeding (18.08%). The least frequent type was intramuscular bleeding (0.70%) (Fig. S1b online).

3.3. Model analysis and the influence of variables on prediction

Pearson correlation tests revealed the correlation between the PLT count at diagnosis and the current low PLT count. Meanwhile, no significant correlation between the other candidate predictors was observed (Fig. S2 online), indicating that the other variables were independent.

In the feature selection process, we applied Lasso regression, which identified 19 features and provided feature importance rankings; the results are summarized in Supplementary Materials. We selected the top ten features for model construction: infection, uncontrolled diabetes mellitus, age, ITP type, cardiovascular disease, low absolute lymphocyte count ($<1 \times 10^9$), skin and mucosa bleeding, initial PLT, low PLT ($<20 \times 10^9$), and disease duration (from highest to lowest; Fig. S3 online). These predictors were used in subsequent modeling processes.

3.4. Performance of the ML models in predicting critical ITP bleeding

We analyzed the predictive performance of different ML models, as shown in Fig. 2. The models demonstrated varied capabilities in predicting critical ITP bleeding (Fig. 2a and Fig. S3, Table S2 online). Out of the seven classification algorithms, RF exhibited the best performance with an AUC of 0.89 for the retrospective cohort (Fig. 2b). We successfully built a simple and accessible web-based application for critical ITP bleeding prediction (https://47.94.162.105:8080/mavenitp/), which was applied to 39 centers located nationwide for prospective testing and exhibited an AUC of 0.82 for inpatients (Fig. 2c). When the model was further extended to outpatients, the AUC was 0.74, indicating a good prediction efficacy (Fig. 2d).

Further, calibration plots confirmed the consistency of the risks of critical bleeding predicted by the RF algorithm with the observed actual counterparts (Fig. 3a). We also conducted DCA to assess the net benefit of the developed prediction model. The analysis revealed that when the probability threshold for the occurrence of critical bleeding was set between 5% and 95%, the net benefit of using the model was significantly superior to that of the "treat none" and "treat all" strategies (Fig. 3b). These findings

Table 1
Clinical characteristics of adult ITP patients with and without critical ITP bleeding.

Characteristics	Training dataset				Internal validation dataset				External test dataset			
	Overall	Without critical ITP bleeding	With critical ITP bleeding	*P* value	Overall	Without critical ITP bleeding	With critical ITP bleeding	*P* value	Overall	Without critical ITP bleeding	With critical ITP bleeding	*P* value
	(*n* = 1494)	(*n* = 1068)	(*n* = 426)		(*n* = 600)	(*n* = 429)	(*n* = 171)		(*n* = 1097)	(*n* = 1047)	(*n* = 50)	
Demographic characteristics												
Age (years), median (Q1, Q3)	54.0 (36.0, 68.0)	54.0 (35.0, 67.0)	53.0 (36.0, 69.0)	0.008[a]	52.0 (37.0, 67.0)	52.0 (36.0, 69.0)	52.0 (38.0, 66.0)	0.851[a]	52.0 (34.0, 65.0)	52.0 (34.0, 66.0)	53.0 (34.0, 63.0)	0.753[a]
Age at diagnosis (years), median (Q1, Q3)	48.0 (30.0, 64.0)	47.0 (29.0, 62.0)	52.0 (33.0, 68.0)	<0.001[a]	48.0 (30.0, 62.0)	49.0 (30.0, 63.0)	48.0 (32.0, 62.0)	0.846[a]	48.00 (30.00, 63.00)	48.0 (30.0, 63.0)	51.00 (33.0, 58.0)	0.688[a]
Male, *n* (%)	564 (37.75)	403 (37.73)	161 (37.79)	0.983[b]	222 (37.00)	149 (34.73)	73 (42.69)	0.068[b]	383 (34.91)	368 (35.15)	15 (30.00)	0.456
ITP type, *n* (%)				0.138[b]				0.827[b]				0.860
Newly diagnosed	542 (36.27)	375 (35.11)	167 (39.20)		241 (40.17)	170 (39.63)	71 (41.52)		522 (47.76)	500 (47.94)	22 (44.00)	
Persistent	253 (16.93)	176 (16.48)	77 (18.08)		91 (15.17)	64 (14.92)	27 (15.79)		140 (12.81)	133 (12.75)	7 (14.00)	
Chronic	699 (46.78)	517 (48.41)	182 (42.72)		268 (44.67)	195 (45.46)	73 (42.69)		431 (39.43)	410 (39.31)	21 (42.00)	
Comorbidities and symptoms, n (%)												
Uncontrolled diabetes	172 (8.13)	113 (7.45)	59 (9.88)	0.065[b]	42 (7.00)	29 (6.76)	13 (7.60)	0.715[b]	52 (4.77)	50 (4.80)	2 (4.08)	0.818[b]
COPD	72 (3.40)	49 (3.23)	23 (3.85)	0.477[b]	20 (3.33)	12 (2.80)	8 (4.68)	0.247[b]	32 (2.93)	30 (2.87)	2 (4.00)	0.644[b]
CVD	143 (6.76)	93 (6.13)	50 (8.38)	0.064[b]	95 (15.83)	57 (13.29)	38 (22.22)	0.007[b]	98 (8.96)	93 (8.91)	5 (10.00)	0.792[b]
CKD	88 (4.16)	52 (3.43)	36 (6.03)	0.070[b]	26 (4.33)	13 (3.03)	13 (7.60)	0.013[b]	42 (3.84)	41 (3.93)	1 (2.00)	0.488[b]
Infection	208 (9.83)	92 (6.07)	116 (19.43)	<0.001[b]	59 (9.83)	22 (5.13)	37 (21.64)	<0.001[b]	143 (13.06)	135 (12.92)	8 (16.00)	0.528[b]
Skin and mucosa bleeding	728 (48.72)	476 (44.57)	252 (59.16)	<0.001[b]	292 (48.67)	203 (47.32)	89 (52.05)	0.296[b]	778 (71.57)	736 (70.97)	42 (84.00)	0.046[b]
Treatments within 1 month, n (%)												
Glucocorticoid	450 (30.12)	295 (27.62)	155 (36.39)	<0.001[b]	292 (48.67)	203 (47.32)	89 (52.05)	0.022[b]	367 (33.55)	352 (33.72)	15 (30.00)	0.587[b]
Danazol	55 (3.68)	30 (2.81)	25 (5.87)	0.005[b]	21 (3.50)	15 (3.50)	6 (3.51)	0.994[b]	24 (2.20)	20 (1.92)	4 (8.00)	0.004[b]
Immunosuppressive agents	97 (6.49)	61 (5.71)	36 (8.45)	0.052[b]	43 (7.17)	28 (6.53)	15 (8.77)	0.036[b]	138 (12.60)	126 (12.06)	12 (24.00)	0.013[b]
IVIg	135 (9.03)	81 (7.58)	54 (12.67)	0.002[b]	55 (9.17)	27 (6.29)	28 (16.37)	<0.001[b]	115 (10.53)	108 (10.37)	7 (14.00)	0.413[b]
Rituximab	22 (1.47)	18 (1.69)	4 (0.94)	0.280[b]	12 (2.00)	7 (1.63)	5 (2.92)	0.037[b]	24 (2.19)	22 (2.11)	2 (4.00)	0.372[b]
Laboratory test results												
PLT count at diagnosis ($\times 10^9$/L), median (Q1, Q3)	22 (7, 49)	27 (9, 50)	17 (5, 40)	<0.001[a]	24 (7, 49)	27 (10, 50)	16 (5, 45)	0.002[a]	13 (5, 29)	13 (5, 29)	8 (4, 25)	0.068[a]
ALC at diagnosis ($\times 10^9$/L), median (Q1, Q3)	1.57 (1.04, 2.18)	1.64 (1.17, 2.20)	1.40 (0.91, 2.11)	<0.001[a]	1.52 (1.06, 2.21)	1.57 (1.18, 2.24)	1.41 (0.87, 2.04)	0.015[a]	1.44 (1.00, 2.02)	1.46 (1.00, 2.03)	1.21 (0.78, 1.81)	0.086[a]
PLT $\leq 20 \times 10^9$/L, *n* (%)	655 (44.28)	431 (40.66)	224 (53.46)	<0.001[b]	271 (45.16)	182 (42.42)	89 (52.04)	0.033[b]	535 (48.76)	496 (47.37)	39 (78.00)	<0.001[b]

ALC: absolute lymphocyte count; CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; CVD: cardiovascular disease; ITP: immune thrombocytopenia; IVIg: intravenous immunoglobulin; PLT: platelet.
[a] Mann-Whitney *U* test.
[b] *Chi*-square test.

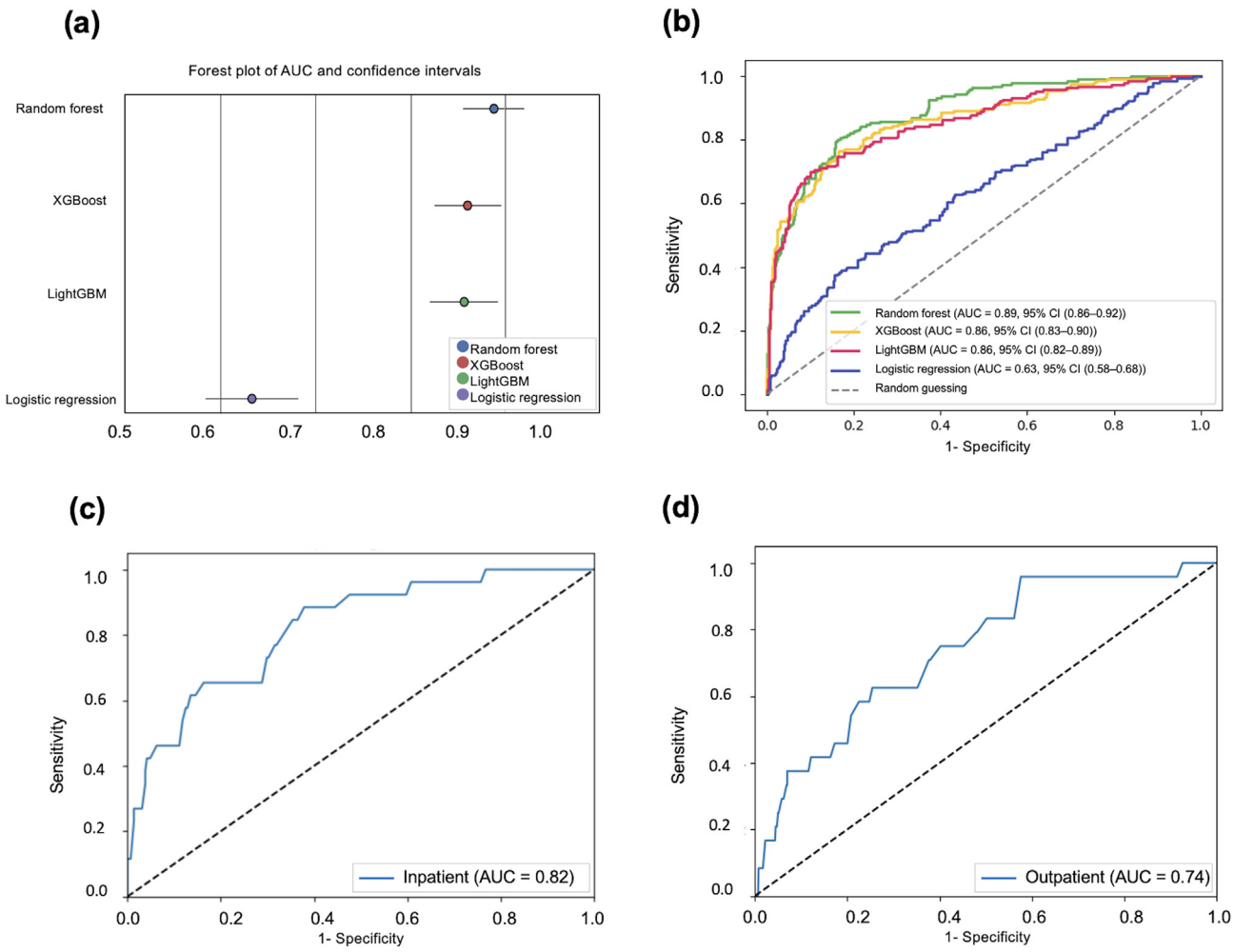

Fig. 2. Performance evaluation of machine learning (ML) models in predicting critical bleeding risk in immune thrombocytopenia (ITP). (a) Forrest plot of the area under the receiver operating characteristic (ROC) curve of different ML models. (b) The ROC curve of different models adopted for predicting the risk of critical ITP bleeding in the retrospective validation cohort. (c) The ROC curve of the random forest (RF) model adopted for predicting the risk of critical ITP bleeding in the prospective inpatient cohort. (d) The ROC curve of the RF model adopted for predicting the risk of critical ITP bleeding in the prospective outpatient cohort.

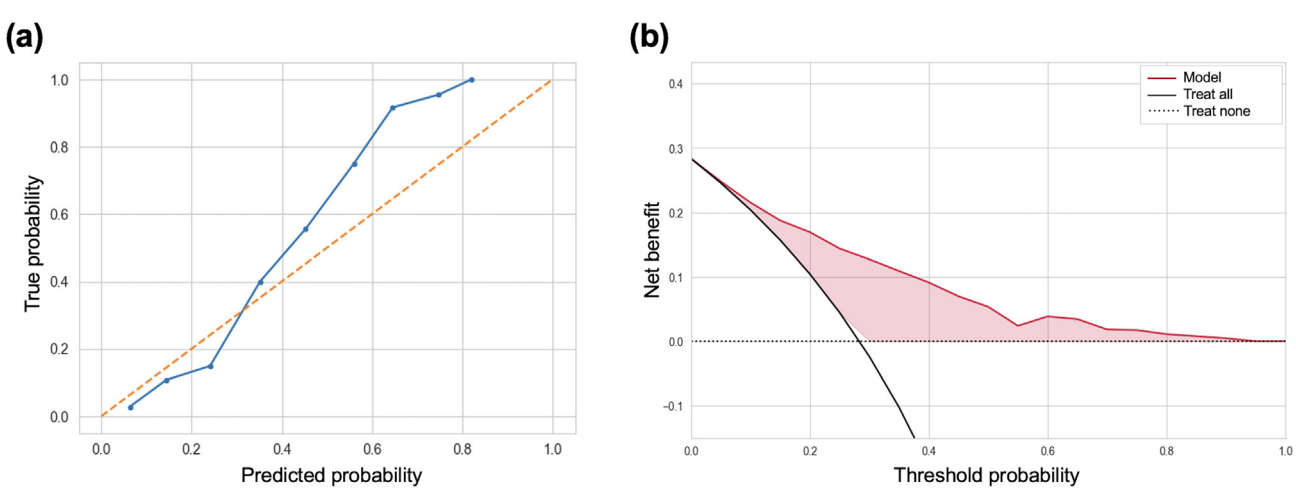

Fig. 3. Evaluation of the calibration and clinical utility of the random forest (RF) model for predicting critical bleeding risks in patients with immune thrombocytopenia (ITP). (a) Calibration plot of the RF model for predicting critical ITP bleeding risks in patients with ITP. The RF model–predicted probability of critical ITP bleeding was plotted on the *x*-axis, and the actual critical ITP bleeding risk was plotted on the *y*-axis. An ideal calibration plot is indicated by a 45° diagonal line. (b) Decision curve analysis of the model. The *x*-axis indicates the threshold probability for critical ITP bleeding. The threshold probability is a level of certainty above which the patient or physician would choose to intervene. The *y*-axis indicates the net benefit, which is calculated as true positive rate − (false positive rate × weighting factor).

indicate that our prediction model has considerable clinical applicability.

3.5. Model interpretation by the Shapley additive explanations method

We further examined the underlying dynamics of our ML model by the Shapley additive explanations (SHAP) method. A SHAP summary plot that depicts the importance of each predictor contributing to the model's prediction of critical bleeding is presented (Fig. S4 online). The plot helps visualize and interpret the complex relationships and interactions between the predictors and outcomes. The predictors are arranged according to their importance, and the color coding represents the magnitude and direction of the effects, providing a clear depiction of the complexity of the prediction model.

4. Discussion

In this study, we applied the newly proposed critical ITP bleeding criteria by ISTH to a large sample size of data and developed a novel and robust ML-based online warning system. The effectiveness of this approach has been demonstrated through multiple applications wherein ML tools outperform proven traditional risk-stratification methods [25–27]. Furthermore, we developed a highly accessible website tool for prediction based on the model. Since all features were derived from routinely collected clinical variables, the developed model can be easily applied to patients admitted with ITP, which can improve clinical decision-making and the identification of groups with high-bleeding risk. We tested the model by prospectively recruiting a testing cohort involving 39 tertiary hematology medical centers across China; the analysis confirmed the performance and robustness of the model.

Critical ITP bleeding is a rare and life-threatening complication during the hospitalization of patients with ITP. The bleeding severely affects the quality of life and increases patient mortality. Although previous studies have identified several predictors, they are inconsistently defined with relatively small sample sizes and lack clinical prediction models. In this study, we employed advanced multiple ML algorithms and compared individual models to identify patients with a high risk of critical ITP bleeding. In addition, we used the most recent ISTH critical ITP bleeding criteria to define severe bleeding. The criteria are based on a review of more than ten different "severe bleeding, major bleeding" definitions, which have been previously proposed and further developed to identify patients with bleeding that require urgent medical treatment [17]. In this study, we developed a model for clinical application to reduce disability and mortality through early interventions for patients with critical ITP bleeding. We implemented the RF algorithm in the web-based online tool to assist clinicians in predicting the risk of bleeding in patients with ITP. Using the most accurate ML model, our online tool aims to facilitate evidence-based decision-making in clinical practice, ultimately leading to improved patient outcomes.

In this study, ICH was found to be the most common site of bleeding, which fulfilled the ISTH critical ITP bleeding criteria, while intraspinal, intraorbital, retroperitoneal, pericardial, and intramuscular compartment syndrome were less frequent [17]. Similar to previous studies [22], a low occurrence of bleeding at these sites was observed, which often starts insidiously and is hence called "silent bleeding". This phenomenon requires specialized physical examination or imaging for detection and is generally severe and rare when clinical signs or symptoms are present [28].

Age, PLT count, infection, uncontrolled diabetes mellitus, and ITP type were identified as predictors through the models. The largest contributor to the model output was the initial diagnosis of PLT count, which reflected the severity of ITP. Age was also a crucial influencing factor; patients with a higher age had a higher risk of bleeding due to comorbidities and vascular degeneration [22]. Furthermore, as previously reported, infections significantly increase the risk of severe bleeding, particularly community-acquired pneumonia [29].

ML extracts complex relationships between multiple variables from high-dimensional large-scale medical data and establishes their associations with outcomes. In recent years, ML has been increasingly applied in medicine and considerably improved the predictive power of various algorithmic models, particularly RF models. Among various ML algorithms for clinical prediction models, the RF model is preferred. It is a supervised ensemble learning model and exhibits superior capabilities in handling high-dimensional multivariate data with nonlinear relationships. Because this study had multiple and complex ITP admission variables with no significant relationships, the RF model was employed to better analyze the data and improve the efficacy of the prediction model. However, the RF model was prone to overfitting, a challenge that was overcome by continuously debugging the model based on a large sample size coupled with independent validation of the prospective cohort [30,31].

While ML has gained considerable attention in generating accurate clinical prediction models, there remains a significant knowledge gap concerning the interpretation of variables derived from these ML "black box" models [32]. This lack of understanding potentially hinders the full exploitation of ML in clinical settings, thus necessitating further exploration and clarification. However, for clinical situations, understanding how the variables affect outcomes, which is a feature that requires good interpretability of the models, is critical. In this study, we adopted the SHAP method for explaining the model. The results were used to understand how the prediction model makes decisions and propose how we can further optimize and modify the model. The SHAP method can provide global and individualized visual interpretation maps that clinicians can use to better understand the role of variables and make confident decisions for their patients [33,34].

Although this study highlights important findings, it should be noted that the algorithm learns from the input features, and some hidden relationships may be lost because of unknown or unregistered features. Furthermore, the data distribution in our study may not be perfectly balanced, leading to potential bias.

In summary, our findings indicate that the ML model utilizing ten readily accessible and straightforward predictors exhibits favorable performance in predicting critical bleeding events in patients with ITP. The development of a web-based online tool using this algorithm has the potential to efficiently assess the bleeding risk profile of patients with ITP, thereby facilitating clinical decision-making and minimizing the occurrence of adversities.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (2021YFC2500300 and 2021YFC2500304), the Key Program of National Natural Science Foundation of China (82230004 and 81730004), the National Natural Science Foundation of China (82200136 and 81970113), ECCM Program of Clinical Research Center of Shandong University (2021SDUCRCB009), and the Capital Health Research and Development of Special Fund (2022-1-4082). We wish to express our sincere gratitude to our esteemed collaborators for their insightful guidance in clinical epidemiology: Prof. Yang Wang from Fuwai

Hospital & Cardiovascular Institute and Professor Si-Yan Zhan from Peking University.

Author contributions

Lei Zhang, Ming Hou, Xiao-Hui Zhang, and Xiao-Jun Huang designed the study. Hu Zhou, Ru Feng, Jian-Ping Shen, Jun Peng, Hai Zhou, Yi Liu, Yong-Ping Song, Lin-Hua Yang, Mei-Yun Fang, Jian-Yong Li, Yun-Feng Cheng, Peng Liu, Ya-Jing Xu, Zhao Wang, Yi Luo, Zhen Cai, Hui Liu, Jing-Wen Wang, Juan Li, Xi Zhang, Zi-Min Sun, Xiao-Yu Zhu, Xin Wang, Rong Fu, Jian-Feng Zhou, Shao-Yuan Wang, Tong-Hua Yang, Li-Ping Su, Liang-Ming Ma, Xie-Qun Chen, Dai-Hong Liu, Hong-Xia Yao, Jia Feng, Hong-Yu Zhang, Ming Jiang, Ze-Ping Zhou, Wen-Sheng Wang, Xu-Liang Shen, Yangji Baima, Yue-Ying Li, Qian-Fei Wang, Tao Wu, Yao-Zhu Pan, Lin Qiu, Da Gao, A-Rong Jin, Wei Li, Su-Jun Gao collected and analyzed data. Qiu-Sha Huang, Hai-Xia Fu, Xiao-Lu Zhu, Yun He, Qian Jiang, Hao Jiang, Jin Lu, Xiang-Yu Zhao, Ying-Jun Chang, assisted in the research. Zhuo-Yu An, Ye-Jun Wu, Yu Hou, Heng Mei, Wei-Xia Nong, Wen-Qian Li wrote the manuscript. All authors critically revised, reviewed, and approved the final version of this manuscript.

Appendix A. Supplementary materials

Supplementary materials to this article can be found online at https://doi.org/10.1016/j.scib.2023.08.001.

References

[1] Kuter DJ, Mathias SD, Rummel M, et al. Health-related quality of life in nonsplenectomized immune thrombocytopenia patients receiving romiplostim or medical standard of care. Am J Hematol 2012;87:558–61.
[2] McMillan R, Bussel JB, George JN, et al. Self-reported health-related quality of life in adults with chronic immune thrombocytopenic purpura. Am J Hematol 2008;83:150–4.
[3] Provan D, Arnold DM, Bussel JB, et al. Updated international consensus report on the investigation and management of primary immune thrombocytopenia. Blood Adv 2019;3:3780–817.
[4] Rodeghiero F, Michel M, Gernsheimer T, et al. Standardization of bleeding assessment in immune thrombocytopenia: report from the international working group. Blood 2013;121:2596–606.
[5] Buchanan GR, Adix L. Grading of hemorrhage in children with idiopathic thrombocytopenic purpura. J Pediatr 2002;141:683–8.
[6] Rosthøj S, Rajantie J, Treutiger I, et al. Duration and morbidity of chronic immune thrombocytopenic purpura in children: five-year follow-up of a nordic cohort. Acta Paediatr 2012;101:761–6.
[7] Robak T, Mainau C, Pyringer B, et al. Efficacy and safety of a new intravenous immunoglobulin 10% formulation (octagam® 10%) in patients with immune thrombocytopenia. Hematology 2010;15:351–9.
[8] Praituan W, Rojnuckarin P. Faster platelet recovery by high-dose dexamethasone compared with standard-dose prednisolone in adult immune thrombocytopenia: a prospective randomized trial. J Thromb Haemost 2009;7:1036–8.
[9] Bennett CM, Rogers ZR, Kinnamon DD, et al. Prospective phase 1/2 study of rituximab in childhood and adolescent chronic immune thrombocytopenic purpura. Blood 2006;107:2639–42.
[10] Bussel JB, Kuter DJ, Pullarkat V, et al. Safety and efficacy of long-term treatment with romiplostim in thrombocytopenic patients with chronic ITP. Blood 2009;113:2161–71.
[11] Bussel JB, Provan D, Shamsi T, et al. Effect of eltrombopag on platelet counts and bleeding during treatment of chronic idiopathic thrombocytopenic purpura: a randomised, double-blind, placebo-controlled trial. Lancet 2009;373:641–8.
[12] Cheng G, Saleh MN, Marcher C, et al. Eltrombopag for management of chronic immune thrombocytopenia (RAISE): a 6-month, randomised, phase 3 study. Lancet 2011;377:393–402.
[13] Kuter DJ, Rummel M, Boccia R, et al. Romiplostim or standard of care in patients with immune thrombocytopenia. N Engl J Med 2010;363:1889–99.
[14] Zhao P, Hou M, Liu Y, et al. Risk stratification and outcomes of intracranial hemorrhage in patients with immune thrombocytopenia under 60 years of age. Platelets 2021;32:633–41.
[15] Radakovich N, Nagy M, Nazha A. Machine learning in haematological malignancies. Lancet Haematol 2020;7:e541–50.
[16] Neunert C, Terrell DR, Arnold DM, et al. American society of hematology 2019 guidelines for immune thrombocytopenia. Blood Adv 2019;3:3829–66.
[17] Sirotich E, Guyatt G, Gabe C, et al. Definition of a critical bleed in patients with immune thrombocytopenia: communication from the ISTH SSC Subcommittee on Platelet Immunology. J Thromb Haemost 2021;19:2082–8.
[18] K/DOQI Workgroup. K/DOQI clinical practice guidelines for cardiovascular disease in dialysis patients. Am J Kidney Dis 2005;45:S1–153.
[19] Loggers SAI, Koedam TWA, Giannakopoulos GF, et al. Definition of hemodynamic stability in blunt trauma patients: a systematic review and assessment amongst Dutch trauma team members. Eur J Trauma Emerg Surg 2017;43:823–33.
[20] Ranieri VM, Rubenfeld GD, Thompson BT, et al. Acute respiratory distress syndrome: the berlin definition. J Am Med Assoc 2012;307:2526–33.
[21] Arnold DM. Bleeding complications in immune thrombocytopenia. Hematology 2015;2015:237–42.
[22] Neunert C, Noroozi N, Norman G, et al. Severe bleeding events in adults and children with primary immune thrombocytopenia: a systematic review. J Thromb Haemost 2015;13:457–64.
[23] Friedman J. Greedy function approximation: a gradient boosting machine. Ann Stat 2001;29.
[24] Moons KG, Altman DG, Reitsma JB, et al. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): explanation and elaboration. Ann Intern Med 2015;162:W1–W73.
[25] Brinkhaus HO, Rajan K, Schaub J, et al. Open data and algorithms for open science in AI-driven molecular informatics. Curr Opin Struct Biol 2023;79:102542.
[26] Hindocha S, Charlton TG, Linton-Reid K, et al. A comparison of machine learning methods for predicting recurrence and death after curative-intent radiotherapy for non-small cell lung cancer: development and validation of multivariable clinical prediction models. EBioMedicine 2022;77:103911.
[27] Walter W, Pohlkamp C, Meggendorfer M, et al. Artificial intelligence in hematological diagnostics: game changer or gadget? Blood Rev 2022:101019.
[28] Tantawy AA, Elsherif NHK, Kenny MA, et al. Silent bleeding in children and adolescents with immune thrombocytopenia: relation to laboratory parameters and health related quality of life. J Thromb Thrombolysis 2020;50:258–66.
[29] Wu YJ, Hou M, Liu HX, et al. A risk score for predicting hospitalization for community-acquired pneumonia in ITP using nationally representative data. Blood Adv 2020;4:5846–57.
[30] Emanuel EJ, Wachter RM. Artificial intelligence in health care: will the value match the hype? J Am Med Assoc 2019;321:2281–2.
[31] Shah NH, Milstein A, Bagley Ph DS. Making machine learning models clinically useful. J Am Med Assoc 2019;322:1351–2.
[32] Wang F, Kaushal R, Khullar D. Should health care demand interpretable artificial intelligence or accept "black box" medicine? Ann Intern Med 2020;172:59–60.
[33] Camacho DM, Collins KM, Powers RK, et al. Next-generation machine learning for biological networks. Cell 2018;173:1581–92.
[34] Ngiam KY, Khor IW. Big data and machine learning algorithms for health-care delivery. Lancet Oncol 2019;20:e262–73.

Zhuo-Yu An is a medical doctoral candidate at Peking University People's Hospital and Peking University Institute of Hematology. His research interest primarily focuses on coagulation disorders and hematopoietic stem cell transplantation, with a particular interest in ITP. He is currently engaged in fundamental, clinical, and translational research related to ITP.

Ye-Jun Wu is a physician and a doctoral candidate at Peking University People's Hospital. Her research interests include hemorrhagic and coagulation diseases and allo-HSCT. Currently, she is engaged in clinical and scientific research on ITP.

Yu Hou is a researcher at the Department of Hematology, Qilu Hospital of Shandong University. She currently serves as a member of the Thrombosis and Hemostasis Group of the Hematology Branch of the Chinese Medical Association and a young member of the Experimental Hematology Professional Committee of the Chinese Society of Pathophysiology. She is engaged in basic and clinical transformation research on molecular immune regulation of primary ITP.

Heng Mei is a professor and vice director at the Institute of Hematology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology. He is the vice director of the Thrombosis and Hemostasis Group, Chinese Society of Hematology. His research interest includes the diagnosis and treatment of complex coagulation disorders and cell therapy.

Wei-Xia Nong is a chief physician and an associate professor at The First Affiliated Hospital of Shihezi University. She worked as a physician for over 20 Years. Her research interest mainly focuses on the diagnosis and treatment of blood system diseases, hematopoietic stem cell transplantation, coagulation diseases, and platelet diseases.

Wen-Qian Li is a chief physician at Qinghai Provincial People's Hospital. She is currently a committee member of the Hematology Branch of the Chinese Medical Association and the chairman of the Hematology Branch of the Qinghai Medical Association. She devoted herself to the diagnosis and treatment of hematological diseases for over 30 years. Her research interest includes the clinical diagnosis and treatment of acute leukemia and the treatment and pathogenesis of bleeding and coagulation diseases.

Lei Zhang is a chief physician and the vice president of the Institute of Hematology & Hospital of Blood Diseases, Chinese Academy of Medical Sciences & Peking Union Medical College. He has long been committed to the clinical diagnosis, treatment, and mechanism exploration of bleeding and coagulation disorders such as hemophilia, myeloproliferative neoplasms, and immune thrombocytopenia. He has proposed new pathogenesis for hematological diseases and developed innovative technologies such as gene therapy and protein drugs for clinical diagnosis and treatment.

Ming Hou is a chief physician at the Department of Hematology, Qilu Hospital of Shandong University. He currently serves as the standing member of the Hematology Branch of the Chinese Medical Association, the standing member of the Experimental Hematology Branch of the Chinese Society of Pathophysiology, and the deputy president of the Hematology Branch of the Chinese Medical Doctor Association. He has been engaged in basic and clinical research on primary ITP for over 30 years.

Xiao-Jun Huang is the director of the National Clinical Research Center for Hematologic Diseases, chairman & chief physician of Peking University People's Hospital & Peking University Institute of Hematology. His research focuses on transplantation immunity. He is the founder of the "Beijing Protocol" for the treatment of leukemia and other hematological diseases.

Xiao-Hui Zhang is the vice director and office director of the National Clinical Research Center for Hematologic Diseases and vice chairman of Peking University Institute of Hematology. He is dedicated to research on hematopoietic stem cell transplantation and coagulation disorders, particularly in overcoming diagnostic and treatment barriers in immune thrombocytopenia.